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CONNECTOR

Abstract

A connector includes: a housing configured to be connectable to a mating connector and forming an assembly space along a connecting direction; and a position assurance member inserted into the assembly space and assembled with respect to the housing to be movable from a temporary locking position to a main locking position along the connecting direction, in which the assembly space is formed between two side walls formed in the housing and extending in the connecting direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)
[0001] The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2024-031917 filed in Japan on Mar. 4, 2024.
BACKGROUND OF THE INVENTION
1. Field of the Invention
[0002] The present invention relates to a connector.
2. Description of the Related Art
[0003] Conventionally, as a connector, for example, as described in Japanese Patent Application Laid-open No. 2017-157454, a connector connected to a mating connector and having a connector position assurance function is known. This connector includes a position assurance member that assures a fitting state with the mating connector. The position assurance member is attached to a housing of the connector and is disposed at a standby position. After the connector and the mating connector are fitted to each other, the position assurance member is moved in the fitting direction, and is moved from the standby position to the fitting lock position. When the position assurance member can move to the fitting lock position, it is assured that the connector and the mating connector are appropriately fitted with each other.
[0004] In the connector described above, there is a portion where the housing and the position assurance member engage, and the mechanical strength of the engaging portion may be lower than that of other portions. Therefore, it is desired to develop a connector capable of improving strength.
SUMMARY OF THE INVENTION
[0005] Therefore, an object of the present invention is to provide a connector capable of improving strength.
[0006] A connector according to one aspect of the invention includes a housing configured to be connectable to a mating connector and forming an assembly space along a connecting direction; and a position assurance member inserted into the assembly space and assembled with respect to the housing to be movable from a temporary locking position to a main locking position along the connecting direction, wherein the assembly space is formed between two side walls formed in the housing and extending in the connecting direction, the housing includes a locking member that engages with the position assurance member assembled in the assembly space and locks the position assurance member at the main locking position, the locking member includes an installation portion installed between the two side walls and provided on a side of a fitting surface facing the mating connector, and the installation portion forms a recess in which a surface facing the mating connector is depressed along the connecting direction.
[0007] The above and other objects, features, advantages and technical and industrial significance of this invention will be better understood by reading the following detailed description of presently preferred embodiments of the invention, when considered in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] FIG. **1** is an exploded perspective view of a connector according to an embodiment;
[0009] FIG. **2** is a perspective view of the connector according to the embodiment;
[0010] FIG. **3** is a plan view of the connector according to the embodiment in an exploded state;
[0011] FIG. **4** is a perspective view of a housing in the connector according to the embodiment;
[0012] FIG. **5** is an explanatory view of a main locking state of a position assurance member in the

connector according to the embodiment;
[0013] FIG. **6** is an explanatory view of connection in the connector according to the embodiment; and
[0014] FIG. **7** is an explanatory view of mechanical strength in the connector according to the embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0015] Hereinafter, an embodiment according to the present invention will be described in detail with reference to the drawings. Note that the present invention is not limited by the embodiment. In addition, constituent elements in the following embodiments include those that can be easily replaced by those skilled in the art or those that are substantially the same.

Embodiment

[0016] The present embodiment relates to a connector. In the following description, among a first direction, a second direction, and a third direction intersecting each other, the first direction is referred to as a “connecting direction X”, the second direction is referred to as a “width direction Y”, and the third direction is referred to as a “height direction Z”. Here, the connecting direction X, the width direction Y, and the height direction Z are orthogonal to each other. The connecting direction X corresponds to a connecting direction or a fitting direction of a connector and a mating connector. The width direction Y corresponds to the width direction of the connector. The width direction Y and the height direction Z correspond to an orthogonal direction orthogonal to the connecting direction X. In addition, each direction used in the following description represents a direction in a state in which each part is assembled to each other unless otherwise specified. Note that the term “orthogonal” as used herein includes substantially orthogonal.
[0017] As illustrated in FIGS. **1** to **3**, a connector **1** according to the present embodiment is a connector used by being connected to a mating connector **10**, and has a connector position assurance (CPA) function. The connector **1** includes a housing **2** and a position assurance member **3**. FIG. **1** illustrates a state in which the connector **1** and the mating connector **10** are spaced apart from each other and the housing **2** and the position assurance member **3** are separated from each other. The housing **2** is a member to be fitted to the mating connector **10**, and is formed in, for example, a block shape. The housing **2** is made of resin, for example, and is formed by resin molding. The housing **2** forms an installation hole **21** for disposing a terminal. The installation hole **21** is formed along the connecting direction X with the mating connector **10**. A plurality of installation holes **21** are formed, and for example, a plurality of installation holes are arrayed in the width direction Y. The installation hole **21** is formed to penetrate the housing **2** in the connecting direction X, and inserts and holds a terminal of a terminal-equipped electric wire (not illustrated).
[0018] In the housing **2**, an assembly space **22** is formed above a region where the installation hole **21** is formed. The assembly space **22** is a space for inserting and assembling the position assurance member **3**, and is formed, for example, between two side walls **23** provided spaced apart from each other in the width direction Y of the housing **2**. The side wall **23** is a wall body provided at each end portion in the width direction Y of the housing **2** and extending in the height direction Z and the connecting direction X. That is, the two side walls **23** partition side portions on both sides of the assembly space **22** to define the assembly space **22**.
[0019] A guide portion **222** is formed in the assembly space **22**. The guide portion **222** is formed along the connecting direction X at a position of a lower end portion of the assembly space **22**. For example, the guide portion **222** is formed to project out in the width direction Y from the inner surface of the side wall **23**. The guide portion **222** guides the position assurance member **3** along the connecting direction X at the time of assembly of the position assurance member **3**. Furthermore, the guide portion **222** functions as a stopper at the time of assembly of the position assurance member **3**. For example, an end face **222**A on a side opposite to a fitting surface **2**A of the guide portion **222** abuts on the position assurance member **3** at the time of assembly of the position assurance member **3**, and arranges the position assurance member **3** at the main locking

position. The fitting surface **2**A is an end face of the housing **2** in the connecting direction X, and is a surface facing the mating connector **10** in a state where the housing **2** is fitted to the mating connector **10**.

[0020] The housing **2** is provided with a locking member **24**. The locking member **24** is a member that locks the housing **2** with respect to the mating connector **10** at the time of connection of the connector **1** and the mating connector **10**, and is a member that engages with the position assurance member **3**. The locking member **24** is provided, for example, in the assembly space **22** and is installed along the connecting direction X.

[0021] The locking member **24** includes a locked portion **241**, a pressing portion **242**, an arm portion **243**, and an installation portion **244**. The locked portion **241** is a site that engages with the locking portion **121** of the mating connector **10** and the locking portion **331** of the position assurance member **3**. The pressing portion **242** is a site that releases the engagement between the locked portion **241** and the locking portion **121** by being pressed and releases the locked state of the mating connector **10** and the housing **2**. Each of the locked portion **241**, the pressing portion **242**, and the installation portion **244** extends in the width direction Y intersecting the connecting direction X, and is connected to, for example, two arm portions **243**. The two arm portions **243** are disposed spaced apart from each other in the width direction Y and extend in the connecting direction X. The locked portion **241**, the pressing portion **242**, and the installation portion **244** are installed between the two arm portions **243**, and the installation portion **244**, the locked portion **241**, and the pressing portion **242** are arranged in this order from the fitting surface **2**A side.

[0022] The locked portion **241** is provided, for example, between the opposing surfaces **243**A of the two arm portions **243**. The pressing portion **242** is provided so as to project out upward in the height direction Z as compared with the locked portion **241**, and is connected to, for example, upper surfaces **243**B of the two arm portions **243**. The installation portion **244** is connected to end portions on the fitting surface **2**A side of the two arm portions **243**. The installation portion **244** is formed to be longer than the distance between the two arm portions **243**, and the end portions on both sides are respectively connected to the side walls **23**. A column portion **25** is connected to a central position of the installation portion **244** in the width direction Y. The column portion **25** is a member extending along the height direction Z, and is provided between the installation portion **244** and a floor portion **221** of the assembly space **22**. The column portion **25** supports the installation portion **244** from below. That is, the locking member **24** is supported at three points on the fitting surface **2**A side, and the strength is increased.

[0023] That is, the locking member **24** is supported by the two side walls **23** and the column portion **25** through the installation portion **244** on the fitting surface **2**A side, and has a structure that is hardly damaged even when receiving an external force. For example, if the installation portion **244** is supported at two points by the two side walls **23**, the installation portion **244** has a structure that is easily bent in the height direction Z, and the strength against the external force decreases. On the other hand, in the connector **1** according to the present embodiment, since the locking member **24** is supported at three points, strength can be improved, and strength against the external force can be enhanced.

[0024] The end portions on the side opposite to the fitting surface **2**A of the two arm portions **243** are connected to the side walls **23**, respectively. That is, the locking member **24** is supported on each of the fitting surface **2**A side and both sides opposite to the fitting surface **2**A. Therefore, the strength of the locking member **24** is improved as compared with the case of the cantilever structure.

[0025] A recess **244**A is formed in the installation portion **244**. The recess **244**A is formed by depressing a surface facing the mating connector **10** in the installation portion **244** along the connecting direction X. For example, the recess **244**A is formed by depressing the entire installation portion **244** in the connecting direction X. Here, the whole includes substantially the whole. The installation portion **244** of the locking member **24** has a structure that is less likely to

receive an external force due to the formation of the recess **244**A. For example, even when the connector **1** falls on the floor, the installation portion **244** is less likely to directly come into abutment with the floor surface, and the locking member **24** is less likely to receive an impact.
[0026] As illustrated in FIGS. **1** to **3**, the position assurance member **3** is configured such that the guide arm **32** and the locking arm **33** project out from a base portion **31** in the connecting direction X. The base portion **31** is a plate-like site raised upward, and is provided in a direction intersecting the connecting direction X. Two guide arms **32** are provided, and extend in the connecting direction X from both end portions in the width direction Y of the base portion **31**. When the position assurance member **3** is assembled to the assembly space **22** of the housing **2**, the guide arm **32** is inserted along the guide portion **222**.
[0027] The locking arm **33** is provided to project out in the connecting direction X between the two guide arms **32**. The locking arm **33** is an engagement portion that engages with the housing **2**. When the position assurance member **3** is assembled to the assembly space **22**, the locking arm **33** is inserted between the two arm portions **243**. The locking arm **33** is formed with a locking portion **331**. The locking portion **331** is a site that engages with the locked portion **241** of the locking member **24** of the housing **2**. When the position assurance member **3** is inserted with respect to the housing **2** up to the temporary locking position (standby position), the locking portion **331** is disposed at a position in front of the locked portion **241**. When the connector **1** is appropriately connected to the mating connector **10**, the position assurance member **3** can be inserted with respect to the housing **2** up to the main locking position, and it can be recognized that the connector **1** is appropriately connected to the mating connector **10**. At this time, due to the insertion of the position assurance member **3**, the locking portion **331** can move toward the fitting surface **2**A side of the housing **2** with respect to the locked portion **241** by going under the locked portion **241**. On the other hand, when the connector **1** is not appropriately connected to the mating connector **10**, the position assurance member **3** cannot be inserted with respect to the housing **2** up to the main locking position. That is, the locking portion **331** cannot move to the fitting surface **2**A side of the locked portion **241** by going under the locked portion **241**.
[0028] As illustrated in FIG. **1**, the mating connector **10** has a tubular shape, and is configured to allow the connector **1** to be inserted into the internal space **12** with the opposing surface **11** opened. A locking portion (not illustrated) is formed on the ceiling surface of the internal space **12**. The locking portion is a protruding portion that engages with the locked portion **241** of the locking member **24**. Since the locked portion **241** is locked to the locking portion, the connector **1** is suppressed from being detached from the mating connector **10**. A plurality of terminals (not illustrated) are provided in the internal space **12**.
[0029] As illustrated in FIG. **4**, the locking member **24** of the housing **2** forms a projection **27** projecting downward from the arm portion **243**. In addition, a protrusion **28** projecting out from the floor portion **221** of the assembly space **22** is formed. Each of the projection **27** and the protrusion **28** is a site that engages with the position assurance member **3**. When the housing **2** is resin molded, the connecting direction X is the direction of mold opening. In this case, when attempting to form the projection **27** and the protrusion **28**, it is necessary to form the hole **26** on the fitting surface **2**A side of the assembly space **22** in order to remove the mold. Although it is conceivable to open the entire portion on the fitting surface **2**A side between the locking member **24** and the floor portion **221**, there is a concern that the supporting strength of the locking member **24** may reduce. Therefore, in the connector **1** according to the present embodiment, the supporting strength of the locking member **24** is increased by providing the column portion **25** to support the locking member **24**.
[0030] As illustrated in FIG. **5**, the position assurance member **3** is assembled to the housing **2** and disposed at the main locking position. At this time, the position assurance member **3** is spaced apart from the locking member **24** and is abutted with the end face **222**A of the guide portion **222**. That is, the position assurance member **3** is inserted from the temporary locking position to the main

locking position in the assembly space **22**, but the abutment surface **311** of the base portion **31** abuts on the guide portion **222** and is disposed at the main locking position. At this time, the position assurance member **3** is spaced apart from the locking member **24**, and a gap G is formed between the position assurance member **3** and the locking member **24**.

[0031] Next, mechanical strength of the connector **1** according to the present embodiment will be described.

[0032] As illustrated in FIG. **6**, the connector **1** is connected to the mating connector **10** by assembling the position assurance member **3** with respect to the housing **2**. The position assurance member **3** is inserted with respect to the housing **2** up to the temporary locking position. An electric wire (not illustrated) is connected to the connector **1** and the mating connector **10**.

[0033] At this time, as illustrated in FIG. **7**, the connector **1** may hit an external object. For example, it is conceivable that the connector **1** accidentally falls and collides with the floor surface **80**. When the connector **1** falls with the fitting surface **2**A facing downward, the fitting surface **2**A collides with the floor surface **80**. In this case, the connector **1** receives an impact from the fall, but application of a large impact to the locking member **24** is suppressed. That is, the recess **244**A is formed in the installation portion **244** of the locking member **24**, and the locking member **24** is provided to be recessed from the fitting surface **2**A. Therefore, in the connector **1** according to the present embodiment, the locking member **24** is suppressed from directly abutting on the floor surface **80**, and the impact received from the floor surface **80** can be reduced.

[0034] In addition, the locking member **24** is supported at three points in the housing **2**, and the mechanical strength is improved. That is, the installation portion **244** of the locking member **24** is connected to the side walls **23** on both sides and is connected to the column portion **25** extending from the floor portion **221** of the assembly space **22**. Therefore, the installation portion **244** has a structure that is less likely to move in the width direction Y and the height direction Z, and has a structure that is less likely to deform when receiving an impact. Therefore, in the connector **1** according to the present embodiment, the locking member **24** is suppressed from being easily damaged even when receiving an impact from the floor surface **80**.

[0035] Furthermore, as illustrated in FIG. **5**, in the connector **1**, when the position assurance member **3** is assembled to the housing **2** and disposed at the main locking position, the position assurance member **3** abuts on the guide portion **222** of the assembly space **22**. At this time, a gap G is formed between the position assurance member **3** and the locking member **24**, and the position assurance member **3** is spaced apart from the locking member **24**. Therefore, even when the connector **1** comes into contact with an external object, an impact is suppressed from being directly applied to the locking member **24**. That is, even when the position assurance member **3** of the connector **1** comes into contact with an external object, an impact is transmitted to the guide portion **222** of the housing **2**, and the impact is less likely to be transmitted to the locking member **24**. In the connector according to the present embodiment, an external force is suppressed from being applied from the locking member **24**, and proof stress against the external force can be improved.

[0036] As described above, the connector **1** according to the present embodiment can suppress an external force from directly applying to the locking member **24** by forming the recess **244**A in the installation portion **244** of the locking member **24**. As a result, the connector **1** according to the present embodiment can reduce the impact applied to the locking member **24**.

[0037] Furthermore, in the connector **1** according to the present embodiment, the position assurance member **3** is spaced apart from the locking member **24** at the main locking position and is abutted on the guide portion **222**. Therefore, when the position assurance member **3** receives an external force, the connector **1** according to the present embodiment can suppress transmission of the external force to the locking member **24** via the position assurance member **3**. Therefore, the connector **1** according to the present embodiment can improve the mechanical strength.

[0038] In addition, in the connector **1** according to the present embodiment, the locking member **24**

is supported at three points in the housing **2**, and thus the mechanical strength of the housing **2** can be enhanced. That is, in the housing **2**, the locking member **24** is supported at three points by being connected to the two side walls **23** and the column portion **25** while having a structure of being installed over the two side walls **23**. Thus, the connector **1** according to the present embodiment can strongly support the locking member **24** and improve the mechanical strength.
[0039] Note that the connector according to the present invention is not limited to the above-described embodiment, and various modifications can be made within the scope described in the claims. The connector according to the present embodiment may be configured by appropriately combining the components of each embodiment and modified examples described above.
[0040] For example, in the above-described embodiment, the case has been described in which the guide portion **222** projects out from the side wall **23** in the locking member **24**, but the guide portion **222** may not be provided, and the position assurance member **3** may be configured to abut on a portion other than the guide portion **222** of the housing **2**. Even in this case, the strength of the connector **1** can be improved by forming the recess **244**A in the installation portion **244** of the locking member **24**.
[0041] According to the connector of the embodiment, strength can be improved.
[0042] Although the invention has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A connector comprising: a housing configured to be connectable to a mating connector and forming an assembly space along a connecting direction; and a position assurance member inserted into the assembly space and assembled with respect to the housing to be movable from a temporary locking position to a main locking position along the connecting direction, wherein the assembly space is formed between two side walls formed in the housing and extending in the connecting direction, the housing includes a locking member that engages with the position assurance member assembled in the assembly space and locks the position assurance member at the main locking position, the locking member includes an installation portion installed between the two side walls and provided on a side of a fitting surface facing the mating connector, and the installation portion forms a recess in which a surface facing the mating connector is depressed along the connecting direction.
2. The connector according to claim 1, wherein the position assurance member is, in a state of being assembled to the housing and disposed at the main locking position, spaced apart from the locking member and abutted on a guide portion formed to project out from the side wall toward a side of the assembly space.